

DEPARTMENT OF MINES
SOUTH AUSTRALIA

GEOLOGICAL SURVEY OF SOUTH AUSTRALIA

MAGNETIC OBSERVATIONS IN THE MIDDLEBACK RANGE AREA
GEOPHYSICAL REPORT 1/53.

by

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Summary

This is a composite of some earlier reports viz., 1/51 and 4/51 together with some additional information and deals with attempts to trace magnetic banded iron formations under alluvial cover.

INTRODUCTION:

During observations by C. Kerr Grant, S. A. Mining Review No. 90, it was established that the steeply dipping iron bearing quartzite formations, common to this area, gave rise to considerable magnetic anomalies. It was then suggested that these could be used to trace such formations in alluvial covered areas.

Under the direction of Dr. K. R. Miles, who was conducting a regional geological survey of this area, observations of the variation of the vertical magnetic intensity were made at intervals along traverses designed to intersect possible buried extensions of iron bearing quartzite formations.

AREAS INVESTIGATED:

The locality plan Fig. 1 shows the areas A,B,C,D,E,F, in which observations were carried out.

Areas A,B,D,E and F are all in the vicinity of the Middleback Range, which runs approximately north and south, some twenty miles west of Whyalla, rising to a maximum of some fifteen hundred feet above sea level.

Most of the terrain surrounding the range is of comparatively low relief being in the main covered by sandy alluvium upon which stunted vegetation, typical of the lower rainfall region of this State, has become established.

Area C is just to the north of Iron Knob, which is about twenty five miles north of the Middleback Range.

GEOLOGY:

The geology of this region is the subject of a bulletin to be issued by this department, and at present in the course of preparation by Dr. K. R. Miles.

Briefly however, the Middleback Ranges are the synclinal portions of much more extensive folded beds, thought to have been deposited in the Middle Pre-Cambrian or Archaean era. Preservation of the range from erosion, which has reduced the level of most of the surrounding terrain, is accredited to the extremely hard nature of the banded iron bearing quartzite, which is a feature of the range.

The banded ferruginous quartzite as previously mentioned, is responsible for the major magnetic anomalies in the area; specimens showing a high percentage of magnetite amongst the ferruginous material forming the dark bands through the lighter quartzite base.

This material is regarded as the source of the iron in the rich hematite deposits held under lease by the Broken Hill Proprietary Company Limited. The structural and genetic associations between the deposits and the quartzites has been discussed in the bulletin in preparation by Dr. Miles.

Running parallel and to the west of the main range a line of low hills which has for reference purposes, been designated the "Camel Hill Line" of hills appears to be the surface expression of one limb of another synclinal fold. Apart from this type of formation exhibiting the banded iron bearing quartzites, most other rock outcrops in the area are, with the exception of some quartz ridges along fault planes, of granitic or gneissic type, and generally occur in the form of small rounded knolls rising only slightly above the surrounding alluvium.

TECHNIQUE

Observations were carried out with a Hilger and Watts vertical magnetic force variometer, stations being established at one tenth mile intervals along traverses whose relative positions were fixed by the use of aerial photographs. Drift control was obtained by the use of the closed loop technique; closures being made after periods of three to four hours. As no change greater than 30 gammas was measured at the repeat stations, errors due to drift should not be of any significance in dealing with these

anomalies of several thousand gammas.

RESULTS

Reference is made to areas A,B,C,D,E,F, the locations of which are shown on Fig. 1. The results obtained in each area are dealt with separately below.

Area A

This area includes the line of low hills referred to previously as the "Camel Hill Line" which had been assumed by some authorities to be erosion residuals of one continuous band of magnetic ferruginous quartzite.

Magnetic evidence presented in the form of profiles of vertical magnetic intensity variations indicates that there are several areas where the supposed continuous line can be crossed without any evidence of the large magnetic anomaly, associated with known outcrops of the iron bearing quartzite material. This is the case in most of the larger gaps between outcrops, most of the smaller gaps showing some evidence of continuity, magnetic anomalies of up to seven thousand gammas being observed.

Area B

As has been mentioned earlier the "Camel Hill Line" is thought to be the surface outcrop of one limb of a synclinal fold, the axis of which by virtue of the westerly dipping outcrop beds should be located west of the Camel Hill. Outcropping to the west however are a number of floors of gneissic granite, which some geologists claim to be outcrops of the basement above which were deposited the Middleback Group of formations, of which the banded quartzites form a part. By implication, therefore, the iron bearing quartzite beds, which dip below the surface in the vicinity of Camel Hill should return to the surface in the alluvial covered area between Camel Hill and the exposed granites.

No magnetic anomalies comparable to those over the outcrops of banded ferruginous quartzite material could be found, and it can therefore be assumed that beds of magnetic ferruginous quartzite do not approach the surface within the area investigated. If the formation does exist, the material must have undergone alteration in such a way as to remove its magnetic properties; e.g. magnetite oxidized to hematite or limonite, or possibly altered

by metasomatic action.

It is thought probable that the trough of low readings extending in a northwest direction from the northern end of the gap in the range may be associated with the end of the magnetic beds, which would thus appear to have been cut off at considerable depth, except on the most northern of this set of traverses, where an anomaly of 1,000 gammas would appear to be associated with a remnant of the formation.

Area C

A northerly extension of the iron bearing quartzite formation associated with the Iron Knob hematite deposits for a distance of about one mile is indicated by an anomaly of 3000 - 4000 gammas. This anomaly, which has the appearance of an anticline pitching downwards to the north, terminates in the vicinity of a fault located on geological evidence between the anomaly and a somewhat smaller magnetic anomaly south of the Corunna Hills, a range geologically younger formations than the banded ferruginous quartzites of the Middleback Ranges.

Area D

This survey was undertaken in an alluvial covered valley which runs approximately in a northwest-southeast direction intersecting the north-south line of the Middleback Range.

The large magnetic anomaly can be associated with the banded iron bearing quartzite formation, outcropping to the north, which may be seen to extend under the alluvial cover, above which the anomaly was mapped.

This anomaly has an abrupt termination just before outcrops of schist on the southern edge of the valley, which could be caused by upward or downward pitching of the formation or termination against a fault. Further investigation possibly including drilling would be necessary to establish the exact cause.

Areas E and F

The Middleback Range consists of a number of keel-shaped formations the major axes of which run north and south. Limbs of these keels appear to pass out into the surrounding plain disappearing under the alluvium, in which the only evidence of underlying

rock is provided by small rounded rises of granitic material.

No evidence of any continuation of the limbs of the keel shaped formations was obtained, suggesting termination of these limbs close to where they disappear under the alluvium.

Additional Work

Some additional work was undertaken over the Ash Range, which contains a form of limonitic chert but no evidence of any significant magnetic anomaly characteristic of the banded ferruginous quartzites was obtained. A small area in the centre of the Kimba Gap was found to exhibit a magnetic anomaly of several thousand gammas but this did not extend far beyond the limits of an outcrop of banded ferruginous quartzite.

CONCLUSIONS

This work has indicated additional factual information of assistance in regional structural mapping of magnetic iron bearing quartzite formations, where no outcrops exist.

Much of the evidence obtained has been negative, but has provided a test of geological hypotheses, and has supplied data useful in planning exploratory drilling.

Associated Drawings.

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23/4/53.



S.A. DEPT. OF MINES

MAGNETOMETER SURVEY

AREA A

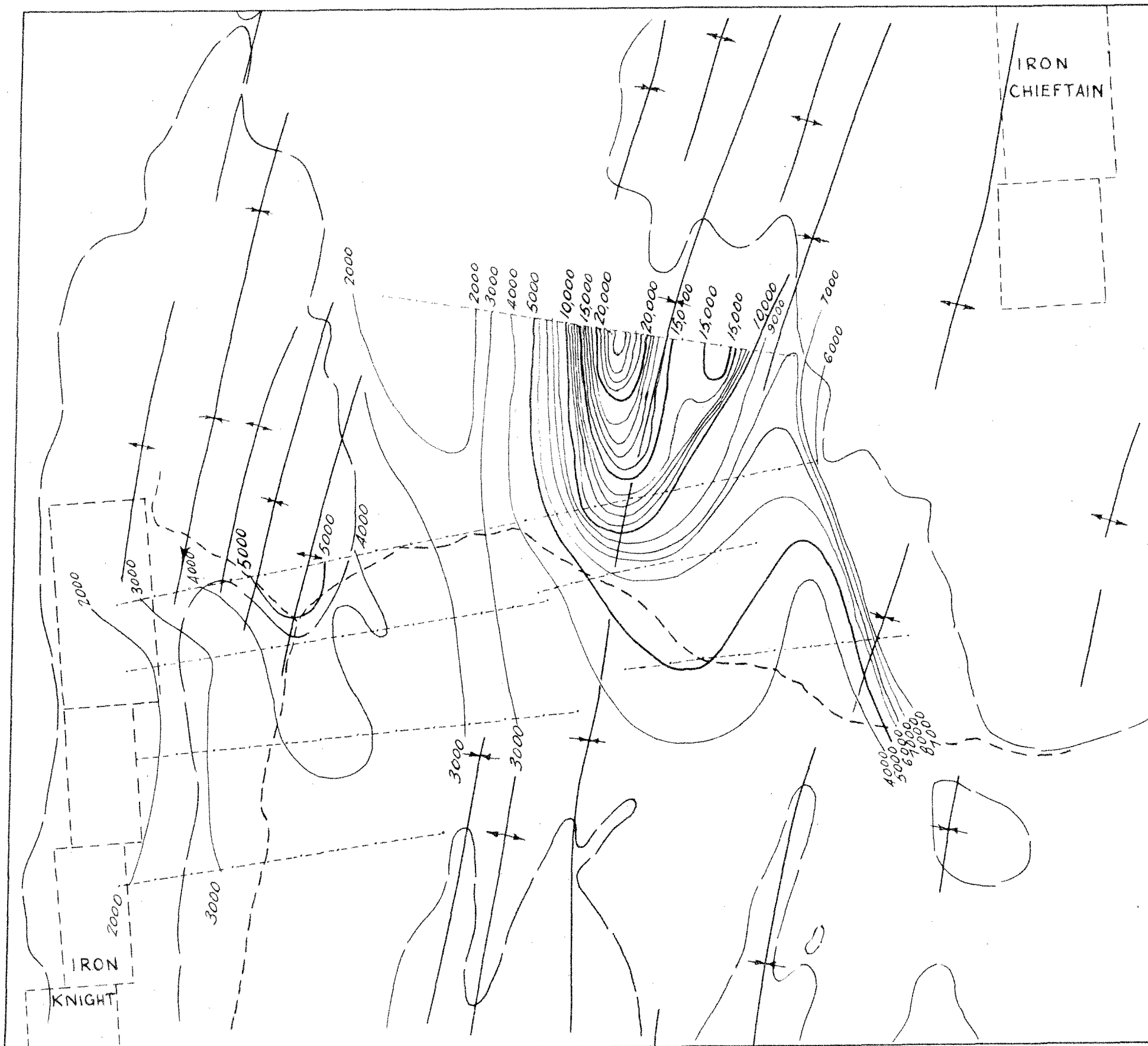
MIDDLEBACK RANGES

Vertical scale 2000 gammas to 1" 40 Chns to 1 inch

53-53 DE 9-3-53

Map No. 53-53
Drawn by
Originals by
D. McPharlin

LINES OF EQUAL VERTICAL MAGNETIC INTENSITY



Contour interval 1000 Gammas

Scale 900 feet to inch

S. A. DEPT. OF MINES

MAGNETOMETER SURVEY
AREA 'D'
MIDDLEBACK RANGE

Approved

Passed

Scale 900' to 1"

Drn. J. McC
Tcd. J. McC
Ckd. R. A.
Exd.

53-46
DE
Date 3-3-53

Director

60

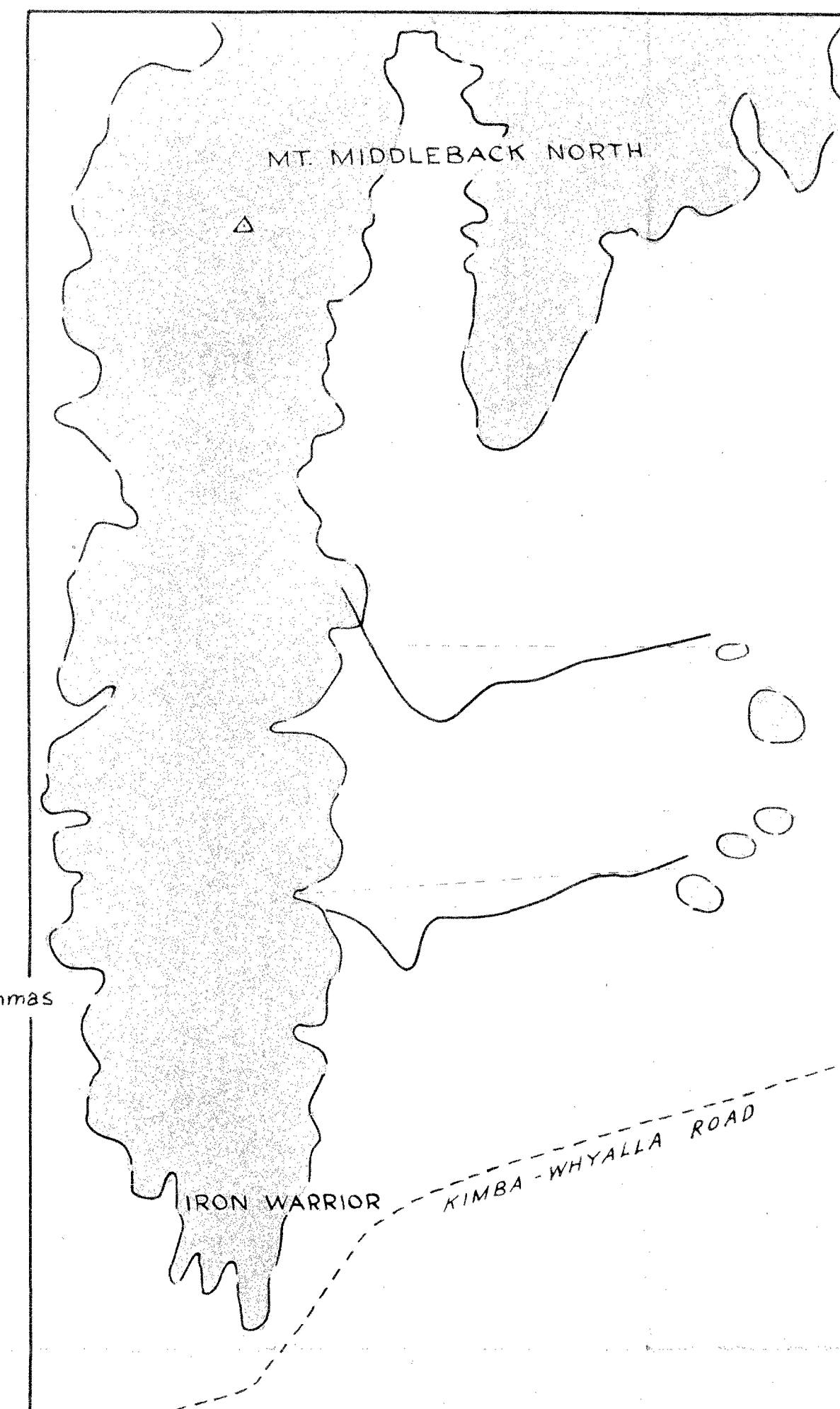
Exd.

Date 3-3-53

No. Amendment Exd Date

AREA E

PROFILES SHOWING VARIATION OF VERTICAL MAGNETIC INTENSITY



SCALE
Horiz. 2" to 1 mile
Vert. 1" to 2000 gammas

AREA F

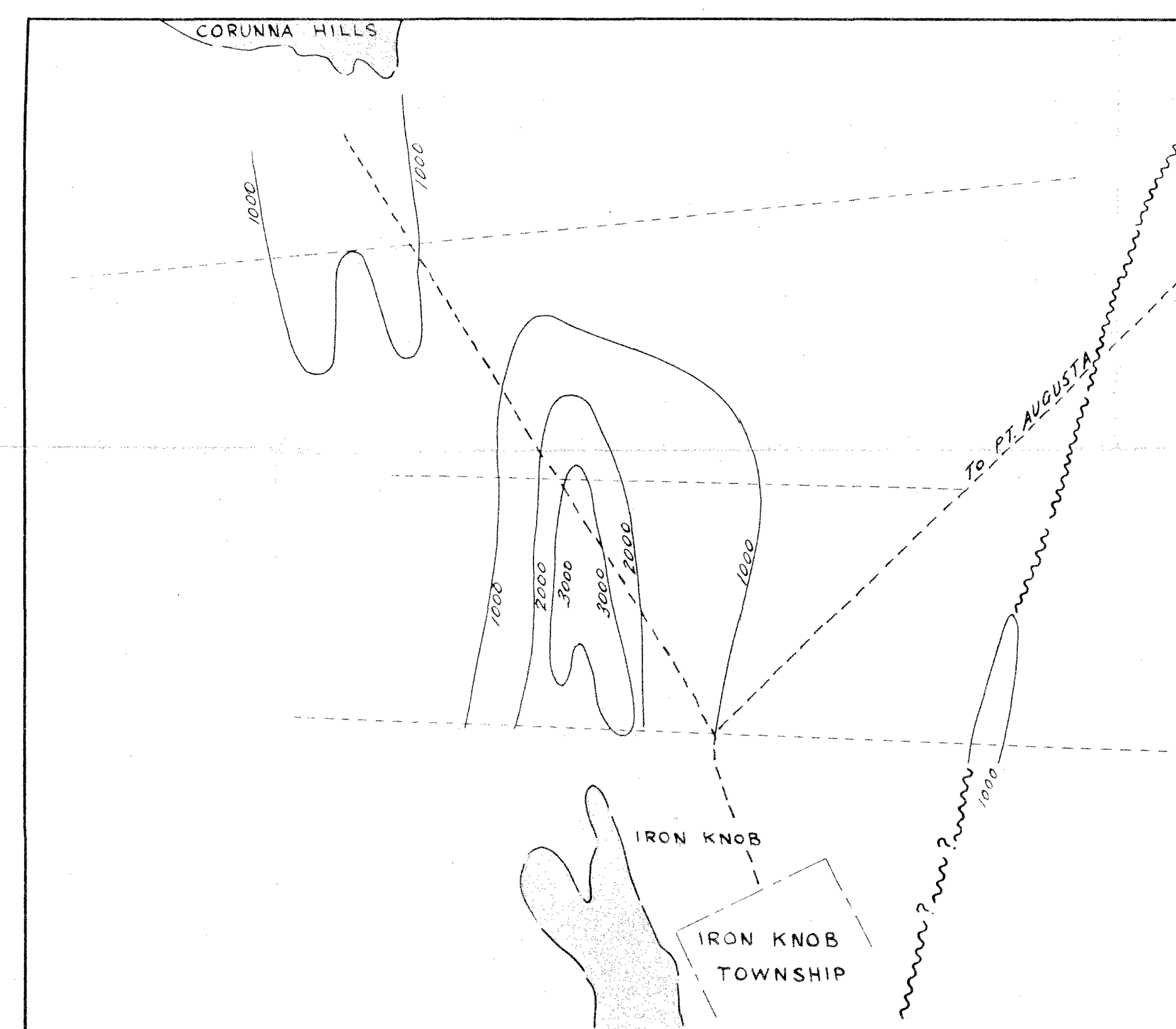
PROFILES SHOWING VARIATION OF VERTICAL MAGNETIC INTENSITY



SCALE
Horiz. 2" to 1 mile
Vert. 1" to 2000 gammas

AREA C IRON KNOB - CORUNNA AREA

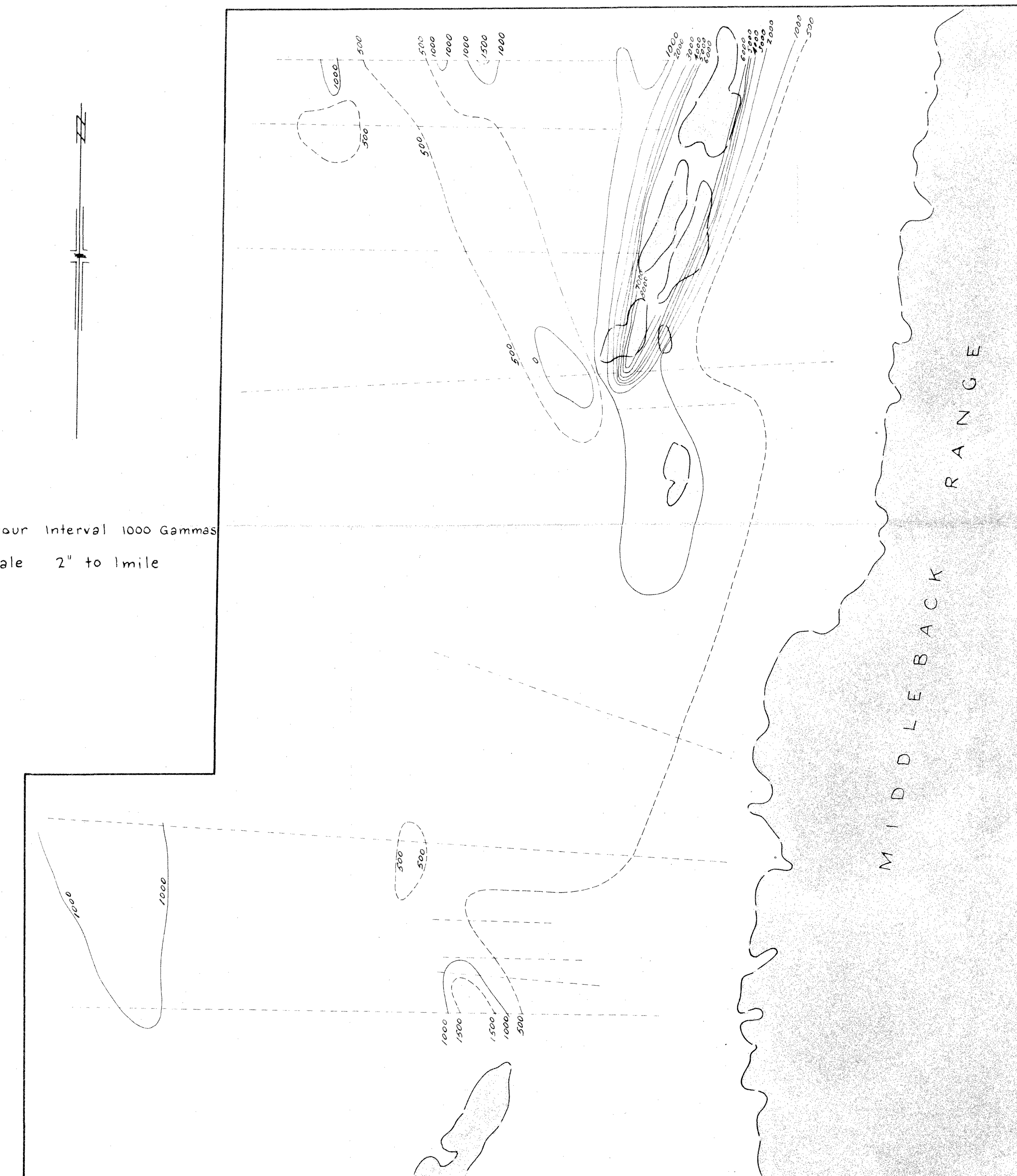
LINES OF EQUAL VERTICAL MAGNETIC INTENSITY



Contour Interval 1000 Gammas
Scale 2" to 1 mile

AREA B

LINES OF EQUAL VERTICAL MAGNETIC INTENSITY



Contour Interval 1000 Gammas
Scale 2" to 1 mile

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MAGNETOMETER SURVEY

AREAS B, C, E & F
MIDDLEBACK RANGES

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Scale 2" to 1 mile

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DE

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